

Abhinav Singh

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SUMMARY

Full-stack Software Engineer building production, cloud-native systems on **Microsoft Azure** with **Java/Spring Boot** and **React/TypeScript**, and applied-AI systems in **Python**. Experienced across backend architecture, event-driven microservices, and LLM integration; comfortable owning features end to end in an **Agile** team, from system design to test coverage and deployment.

EXPERIENCE

Software Engineer

2025 – Present

Bajaj Finserv Health (BFHL)

Pune, India

- Built an automated mailer system in **Java/Spring Boot** that queries and processes issuance and endorsement data from **MongoDB** to generate consolidated failure reports and notify external partners directly, cutting manual reconciliation effort and giving partners faster visibility into their failures.
- Designed and built the **Monitor Hub** observability subsystem tracking issuance, SSO, and communications pipelines — pluggable data collectors, scheduled snapshots, deviation reports, and two-hourly failure alerts — with unit tests across collectors, scheduler, and repositories.
- Implemented a **Salesforce circuit breaker with DLQ replay** for policy issuance, adding resilient retry and recovery to a Kafka-driven event flow.

PROJECTS

OmniMemory | *Python, SQLite, tree-sitter, D3.js, Claude Code Plugin* | github.com/SinghAbhinav04/Omni-Memory 2026

- Built a branch-aware, git-anchored **memory and context layer for AI coding agents** (Claude Code, Antigravity) with a zero-dependency Python/SQLite core requiring no API key; published as a pip package and Claude Code plugin with 5,000+ downloads.
- Added **cross-IDE, cross-device, and team-shared memory sync** alongside a notes-style local dashboard, letting a memory captured in one editor or machine surface for the same user or teammates elsewhere, plus a browsable knowledge graph and repo/branch graph.
- Designed a context-aware **BM25F ranking engine** (symbol/file/prose field weighting) that boosts memories anchored near the code being edited via a call-graph and reinforces ones the agent has previously cited, with no embeddings or LLM calls required at retrieval time.
- Built **symbol-level staleness detection** using a tree-sitter code graph (Python/JS/TS) to flag memories only when their underlying symbol changes in git, plus automated hygiene that quarantines memories from abandoned or merged branches; wired capture/injection through deterministic session hooks rather than relying on agent self-reporting.

Mixture-of-Recursions (MoR) | *Python, PyTorch* | github.com/SinghAbhinav04/Mixture.Of.Recursions

2026

- Implemented an adaptive-depth **language model** from scratch in **PyTorch** that routes each token through a variable number of layers, cutting compute on easy tokens.
- Built a LLaMA-style decoder (RMSNorm, RoPE, SwiGLU, grouped-query attention, Flash Attention) with expert- and token-choice routing and recursive KV caching for inference.
- Achieved up to **2× faster inference** and **40–60% fewer FLOPs** with $\sim 3\times$ fewer unique parameters via cross-recursion weight sharing, at comparable modeling quality.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, HTML/CSS

Backend: Spring Boot, Spring AI, REST APIs, Apache Kafka, Microservices, Event-Driven Architecture

Frontend: React, Redux, Vite, Node.js, Material-UI

AI/ML: PyTorch, LLM Integration, LiteLLM, RAG, Prompt Engineering, Transformers

Data & Cloud: MongoDB, Redis, PostgreSQL, Microsoft Azure, Azure Pipelines, Docker

Practices & Tools: Software Architecture, Agile Methodologies, System Design, Git, CI/CD, Unit Testing

EDUCATION

Chitkara University

Bachelor of Technology in Computer Science

Baddi, India

Aug. 2023 – May 2027